

# YIN Haoteng 殷浩騰

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## EDUCATION

- **Purdue University** West Lafayette, IN  
• **Ph.D. in Machine Learning (CIGP)**; Department of Computer Science 2019 – 2024 (*Expected*)
- **Peking University** Beijing, P.R. China  
• **MPhil in Data Science**; Academy for Advanced Interdisciplinary Studies 2016 – 2019
- **Shandong University** Ji’nan, P.R. China  
• **BEng in Computer Science and Technology**; Taishan College\* (Honors School) 2012 – 2016  
• **BEcon in Finance**; School of Economics 2014 – 2016  
\* Top-notch Undergraduate Training Program by Ministry of Education, P.R. China

## RESEARCH INTERESTS

- Machine Learning and Computation on Graphs  
Topics in Scalable and Parallel Algorithms on Graphs; Trustworthy ML (Data Privacy with LLMs, Fairness in Ranking); Algorithm System Co-design; Spatiotemporal Data Analysis.

## PUBLICATIONS & MANUSCRIPTS (ASTERISK \* INDICATES EQUAL CONTRIBUTIONS)

- [C1] Tianyi Zhang\*, **Haoteng Yin\***, Rongzhe Wei, Pan Li, Anshumali Shrivastava. Learning Scalable Structural Link Representations with Bloom Signatures. *Proceedings of the ACM Web Conference*. (**WWW’24**)
- [C2] Wei Xiang\*, **Haoteng Yin\***, He Wang, Xiaogang Jin. SocialCVAE: Predicting Pedestrian Trajectory via Interaction Conditioned Latents. *The 38th Annual AAAI Conference on Artificial Intelligence* (**AAAI’24**).
- [C3] **Haoteng Yin**, Muhan Zhang, Jianguo Wang, Pan Li. SUREL+: Moving from Walks to Sets for Scalable Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment* 16(11): 2939-2948. DOI:10.14778/3611479.3611499 (**VLDB’23**)
- [C4] Rongzhe Wei, **Haoteng Yin**, Junteng Jia, Austin R. Benson, Pan Li. Understanding Non-linearity in Graph Neural Networks from the Bayesian-Inference Perspective. *Advances in Neural Information Processing Systems* 35 (**NeurIPS’22**)
- [C5] **Haoteng Yin**, Muhan Zhang, Yanbang Wang, Jianguo Wang, Pan Li. Algorithm and System Co-design for Efficient Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment* 15(11): 2788-2796. DOI: 10.14778/3551793.3551831 (**VLDB’22**)
- [C6] Haorui Wang, **Haoteng Yin**, Muhan Zhang, Pan Li. Equivariant and Stable Positional Encoding for More Powerful Graph Neural Networks. *International Conference on Learning Representations* (**ICLR’22**)
- [C7] Bing Yu\*, **Haoteng Yin\***, Zhanxing Zhu. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *The 27th International Joint Conference on Artificial Intelligence* (**IJCAI’18**).
- [J1] Rongzhe Wei, Eleonora Kreačić, Haoyu Wang, **Haoteng Yin**, Eli Chien, Vamsi K Potluru, Pan Li. On the Inherent Privacy Properties of Discrete Denoising Diffusion Models. *Transactions on Machine Learning Research* (**TMLR’24**)
- [J2] **YIN Haoteng** and LIU Yang. Semantic analysis of spatial temporal trajectory in LBSNs. (in Chinese) *SCIENTIA SINICA Informationis* 47(8), 1051-1065(2017). DOI: 10.1360/N112016-00310 (**SMP’16 Best Paper Award**)
- [W2] Prasita Mukherjee, **Haoteng Yin**. OCTAL: Graph Representation Learning for LTL Model Checking. *The 10th International Workshop on Deep Learning on Graphs* (**DLG-KDD’23**, Contributed Talk)
- [W3] **Haoteng Yin**, Yanbang Wang, Pan Li. Revisiting graph neural networks and distance encoding from a practical view. *The 5th International Workshop on Deep Learning on Graphs* (**DLG-AAAI’21**)
- [T1] **Haoteng Yin**, Beatrice Bevilacqua, Bruno Ribeiro, Pan Li. Enhancing Graph Representation Learning through Subgraph Strategies. *The 33rd International Joint Conference on Artificial Intelligence* (**IJCAI’24**, Tutorial Track).
- [TR1] Bing Yu\*, **Haoteng Yin\***, Zhanxing Zhu. ST-UNet: A Spatio-Temporal U-Network for Graph-structured Time Series Modeling. *arXiv preprint arXiv:1903.05631* (Technical Report)

## INVITED TALKS & PRESENTATIONS

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- (2024) Towards Representation Learning on (Sub)Graphs. *Purdue Database Group Seminar*, W. Lafayette, IN
- (2024) Algorithm and System Co-design for Scalable and Efficient Subgraph-based Graph Representation Learning. *Learning on Graphs & Geometry Reading Group*, Online [Link][Video]
- (2023) SUREL+: Moving from Walks to Sets for Scalable Subgraph-based Graph Representation Learning. *The 2nd RCAC Annual Cyberinfrastructure Symposium, CI + AI 2023*, W. Lafayette, IN
- (2023) Towards Scalable and Efficient Subgraph-based Graph Representation Learning. *Amazon A9 Machine Learning Reading Group*, Palo Alto, CA
- (2019) Spatio-Temporal Graph Prediction through Deep Graph Neural Networks. *The 3rd Peking University Graduate Student Forum on Advanced Interdisciplinary Studies*, Beijing, China [Video][**Best Oral Presentation Award**]
- (2018) Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *The 2nd Peking University Graduate Student Forum on Advanced Interdisciplinary Studies*, Beijing, China

## PROFESSIONAL SERVICES

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- Conference Reviewer
  - ICLR'24, ICML'22-24, NeurIPS'22-24, WWW'22,24, WSDM'23, LoG'22-23, CISS'23, IEEE SPAWC'21
- Journal Reviewer
  - **Nature** Machine Intelligence; **ACM** Transactions on Spatial Algorithms and Systems; **IEEE** Transactions on Big Data, Cybernetics, Emerging Topics in Computing, Information Forensics & Security, Intelligent Transportation Systems, Signal Processing
- Graduate Teaching Assistant
  - CS242 Introduction to Data Science - 19'Fall
  - CS38003 Python Programming - 20'Spring, 21'Spring, 23'Spring
  - CS490-LDA Large Scale Data Analytics - 20'Fall
  - CS590-DE I&II Data Engineering - 20'Spring, 20'Summer
  - CS590-NCDS Numerical Computing for Data Science - 21'Spring
- Poster Session Judge - Purdue Spring Undergraduate Research Conference *Apr. 2024*
- Student Volunteer - The International Machine Learning Society *Jul. 2018*
- Student Membership - AAI, ACM, SIAM

## RESEARCH EXPERIENCE

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- **Purdue University** West Lafayette, IN  
*Graduate Assistantship* *Aug. 2019 - Current*
  - **Differentially Private Graph Learning for Foundation Models**  
Research Project (supported by GCoM@CS, Purdue), under the supervision of Prof. Pan Li.
  - **Efficient and Scalable Subgraph-based Graph Representation Learning (SGRL)**  
Research Project (supported by GCoM@CS, Purdue & JPMorgan Chase), under the supervision of Prof. Pan Li.  
Design novel frameworks SUREL/SUREL+ for scalable subgraph-based GRL through algorithm and system co-design. A decoupled paradigm is proposed to learn the representation of walk- and set-based subgraphs, which breaks the computational bottleneck and enables efficient training and inference for large-scale link, relation, and higher-order prediction tasks. SUREL/SUREL+ ranks in the Top 5 on Open Graph Benchmark datasets.
- **LIX, École Polytechnique & Inria** Palaiseau, France  
*Research Intern* *Jun. 2018 - Sept. 2018*
  - **Machine Learning Attacks to Location Privacy**  
Research Project (supported by team Comète, EP), under the supervision of Prof. Catuscia Palamidessi.  
Developed a model of adversary that uses machine learning algorithms to capture users' movement patterns from perturbed trajectory data, which are gathered from location-based services and protected by a privacy mechanism, and then performs inference attacks to measure the vulnerability of the mechanism.

- **Center for Data Science, Peking University** Beijing, P.R. China  
*Graduate Research Assistant* *Jan. 2017 - Aug. 2019*
  - **Traffic Prediction with Deep Spatial Temporal Graph Neural Networks**  
Research Project (supported by Deep Learning Lab, PKU), under the supervision of Prof. Zhanxing Zhu.  
Design a deep learning framework STGCN to predict traffic time series, which coherently models the topology of road networks and captures periodic traffic patterns through spatiotemporal convolutional blocks. It has been widely used as a SOTA graph machine-learning model for transportation research (over 3,600 citations).

## INDUSTRIAL EXPERIENCE

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- **Amazon Search** Palo Alto, CA  
*Applied Scientist Intern* *Jun. 2023 - Sep. 2023*
  - **Graph Foundation Models for Large-scale Text-attributed E-commerce Network**  
Mentors: Shweta Garg and Dr. Changhe Yuan.

## TECHNICAL SKILLS

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- **Languages:** Chinese (native), English (professional working proficiency, OEPT certified)
- **Programming:** Proficient in Python/C, PyTorch, HuggingFace, Opacus, PyG, OpenMP, L<sup>A</sup>T<sub>E</sub>X, Git  
Experienced in C++, PySpark, TensorFlow, Java, Julia, R, MATLAB

## SELECTED HONORS & AWARDS

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- Purdue Teaching Academy Graduate Teaching Award (2%) *Apr. 2024*
- AMiner AI 2000 Most Influential Scholar Award (Honorable Mention) in AAAI/IJCAI *Mar. 2024*
- The Top Reviewer Award (8%) for NeurIPS'22 *Oct. 2022*
- Graduate School Summer Research Grants, Purdue University *Mar. 2021*
- Academic Excellence Scholarship, Peking University (~8%) *Oct. 2016, 2017, 2018*
- The 2nd DiDi-IEEE Elite Forum Extraordinary Potential Award (2/39) *May. 2018*
- École Polytechnique International Student (Intern) Fellowship *Mar. 2018*
- The Most Technical Difficulty Award (Team Leader), Schlumberger HackPKU *May. 2017*
- Guanghua-Yintai PKU Social Innovation DO Camp Scholarship (€1,600) *Jan. 2017*
- The 5th National Conference of Social Media Processing the Best Paper Award (1/25) *Oct. 2016*
- Taishan College International Study Scholarship (¥33,000) *Jun. 2015*